

Caching Concepts

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Learning Objectives

Be able to...

- Explain why caching accelerates computer systems
- Define a working set and identify it in an access trace
- Distinguish between temporal and spacial locality

What is a cache?

- Suppose you run a program that occasionally reads from disk and mostly works in memory. What part *feels* slow?
 - The disk access!
 - [Why: accessing disk is 1,000 slower than referencing memory](#)
- Cache: small, fast storage that keeps a subset of data from a larger, slow storage to reduce average access time.
 - Within a CPU: L1/L2 cache hides DRAM latency
 - OS page cache hides disk latency
 - Browser cache hides network latency
 - Others?
- Terminology:
 - Hit: when an access occurs and data is in the cache
 - Miss: when an access occurs and data is not in the cache
 - Average memory access time (AMAT)
 - For a single-level cache: $\text{HitRate} \times \text{HitTime} + \text{MissRate} \times \text{MissTime}$
 - E.g., hittime = 1, misstime = 100 cycles, rates: 10% and 1%
 - Upshot: small reduction in miss rate makes huge performance gains
- How big should the cache be?